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Project Synopsis

Title : PDF Redaction System — Automated Sensitive Data Detection and Removal

Introduction :

In today's digital world, sensitive personal information such as names, dates of birth, Aadhaar numbers, PAN cards, phone numbers, and email addresses are frequently present in documents shared across organizations. Unauthorized exposure of this data leads to privacy violations and legal non-compliance. There is a growing need for an automated system that can detect and remove such information efficiently.

This project proposes an AI-powered PDF Redaction System that automatically detects sensitive data in uploaded PDF documents and applies redaction by replacing the identified content with [REDACTED] text. The system includes a page-by-page preview, batch processing, custom rule profiles, and a complete audit trail — all running locally without any internet dependency.

Problem statement :

Organizations routinely share PDF documents containing personally identifiable information (PII) without proper redaction. Manual redaction is time-consuming, error-prone, and inconsistent. Existing tools are either cloud-based (creating privacy risks) or require expensive licenses. There is a need for a fully local, intelligent redaction system that can accurately identify and remove sensitive data while preserving the original document format and structure.

Objectives :

The main objectives of this project are:

- To automatically detect sensitive information using NLP, regex, and AI-based models
- To provide a page-by-page PDF preview with highlighted sensitive regions before redaction
- To support batch processing of multiple PDF files with ZIP download
- To allow users to save and reuse custom redaction profiles for different document types
- To implement automatic deletion of uploaded files after redaction for privacy protection
- To maintain a complete audit log of all redaction operations
- To support scanned PDFs using OCR (Tesseract) for text extraction
- To package the system as a standalone desktop application using Electron

Proposed System :

The system works as follows:

1. User uploads a PDF file (single or multiple) through the web interface
2. Backend extracts text and word-level coordinates using PyMuPDF
3. If no text found, OCR engine (Tesseract) processes the scanned pages

4. Detection engine (Presidio + regex) identifies sensitive entities with confidence scores
5. Page preview is generated with red highlight boxes over detected sensitive areas
6. User reviews highlights and confirms redaction
7. Redactions are applied and output PDF is saved
8. Original uploaded file is automatically deleted from the server
9. Audit log records all redaction activity in SQLite database

Technologies Used :

- Python (FastAPI, PyMuPDF, Presidio, Tesseract, SQLite)
- React.js + Vite (Frontend)
- spaCy / Microsoft Presidio (NLP Entity Recognition)
- Tesseract OCR + pdf2image (Scanned PDF Support)
- Electron.js (Desktop Application Packaging)
- SQLite (Audit Log Database)
- Regex Engine (Pattern-based Detection)

System Architecture :

Input → PDF Upload (Single / Batch)
↓
Text Extraction (PyMuPDF / Tesseract OCR)
↓
Detection Engine (Presidio + Regex + Custom Rules)
↓
Preview Generation (Page Images + Highlight Coordinates)
↓
User Review + Confirmation
↓
Redaction + Auto-Delete + Audit Log
↓
Output → Redacted PDF Download (Single / ZIP)

Expected Output :

1) Accurate detection of sensitive PII data including names, dates of birth, Aadhaar, PAN, phone numbers, and email addresses.

2) Page-by-page visual preview with red highlight boxes showing exactly what will be redacted before confirmation.

3) Clean redacted PDF output with [REDACTED] labels, preserving original document layout and structure.

4) Batch redaction of multiple PDFs with ZIP download for organizational use.

5) Saved redaction profiles for quick reuse across different document types such as HR, medical, and legal.

6) Automatic deletion of uploaded files after processing and complete audit trail maintained in a local database.

7) Full support for scanned PDFs through integrated OCR processing.

8) Standalone desktop application (.exe) for easy local deployment without any internet requirement.